1. What is Logistic Regression and why is it suitable for classification?

Logistic Regression is a supervised learning algorithm used for binary classification. It predicts probabilities using the sigmoid function and maps outputs between 0 and 1. It is suitable because it provides clear decision boundaries and interpretable results.

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2. Explain classification performance metrics and why accuracy alone is insufficient.

Classification metrics include accuracy, precision, recall, and F1-score. Accuracy measures overall correctness but can be misleading in imbalanced datasets. A model may predict the majority class well while completely ignoring the minority class.

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3. Define Type-I Error and Type-II Error in the context of risk prediction.

Type-I Error (False Positive): Predicting high risk when the customer is actually low risk.
Type-II Error (False Negative): Predicting low risk when the customer is actually high risk.
In risk prediction, Type-II errors are often more dangerous.

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4. Explain Precision, Recall, F1-Score, TPR, and FPR.

Precision: Correct positive predictions out of total predicted positives.
Recall (TPR): Correct positives out of actual positives.
F1-Score: Harmonic mean of precision and recall.
FPR: Incorrect positive predictions out of actual negatives.

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5. What is AUC-ROC and how does it help in evaluating classifiers?

AUC-ROC measures a model's ability to distinguish between classes. ROC curve plots TPR vs FPR at different thresholds. Higher AUC means better model performance across all classification thresholds.

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6. Why does imbalanced data create problems in classification models?

Imbalanced data causes models to favor the majority class, leading to poor detection of the minority class. This results in misleading high accuracy but low recall for important cases (e.g., high-risk customers).
